

NYTRO® 11GBX-US

Electrical insulating oil

Nytro 11GBX-US is a type II ASTM D3487-16 transformer oil. Developed and formulated to deliver solid resistance to oil degradation, Nytro 11GBX-US provides good oxidation stability for a longer transformer life with less maintenance.

Designed for heavy duty

This product has been specially developed for use in oil-filled electrical equipment – including power and distribution transformers, rectifiers, circuit breakers and switchgears.

Performance and benefits

Good heat transfer. Thanks to low viscosity and viscosity index, this high grade offers extremely good heat transfer characteristics, ensuring heat is efficiently removed from core and windings.

Reliable oxidation stability. Developed and formulated to deliver good resistance to oil degradation, this grade also provides good oxidation stability for enhanced transformer life and minimum maintenance.

Very good low temperature properties. Naphthenic characteristics allow the transformer to start at the lowest possible temperature – without using pour point depressants.

High dielectric strength. This insulating oil both meets and exceeds the toughest demands on dielectric strength – when stored and handled correctly.

Product description

Nytro 11GBX-US is an inhibited oil and fulfils the requirements for ASTM D 3487-16 Type II. Nynas classify this product as a high grade.

Nytro 11GBX-US is rigorously analysed and passes the following corrosion tests:

- ASTM D1275
- IEC 62535
- DIN 51353

PCB: Not detectable according to ASTM D 4059
DBDS: Not detectable according to method using GC-AED.

There's more to us than this

We're delighted you chose one of our transformer oils. If you have any questions about other products and services, get in touch with your local Nynas contact. Besides top quality oils, we offer a wide range of services, including rapid delivery worldwide, sample analysis, training, seminars and much more. All you have to do is ask. Find out more at www.nynas.com

NYTRO[®] 11GBX-US

Property	Unit	Test Method	Specification Limits		Typical Data
			Min	Max	
1 - Physical					
Aniline Point	°C	ASTM D611	63		77
Colour		ASTM D1500		0.5	< 0.5
Flash Point, COC	°C	ASTM D92	145		158
Interfacial tension at 25°C	mN/m	ASTM D971	40		48
Pour point	°C	ASTM D97 / ASTM D5950		-40	-48
Relative Density at 15°C/15°C		ASTM D 1298 / D 4052		0.91	0.882
Viscosity, 100°C	mm²/s (cSt)	ASTM D445		3	2.4
Viscosity, 40°C	mm²/s (cSt)	ASTM D445		12	9.2
Viscosity, 0°C	mm²/s (cSt)	ASTM D445		76	53
Visual Examination		ASTM D1524	Clear and bright		Clear and bright
2 - Electrical					
Breakdown voltage at 60 Hz		ASTM D1816			
- VDE electrodes, As received	kV	1 mm gap	20		> 40
- VDE electrodes, As received	kV	2 mm gap	35		> 46
- VDE electrodes, As processed*	kV	2 mm gap			> 70
Dielectric breakdown voltage, impulse conditions, negative polarity point	kV	ASTM D3300	145		> 300
Gassing tendency, Hydrogen	µL/min	ASTM D2300		30	14
Dielectric dissipation factor at 60 Hz		ASTM D924			
DDF at 100°C	%	ASTM D924		0.3	< 0.1
DDF at 25°C	%	ASTM D924		0.05	0.001
3 - Chemical					
Oxidation stability at 110°C		ASTM D2440			
- 72 h, sludge	wt %	ASTM D2440		0.1	0.01
- 72 h, total acid number	mg KOH/g	ASTM D2440		0.3	0.04
- 164 h, sludge	wt %	ASTM D2440		0.2	0.02
- 164 h, total acid number	mg KOH/g	ASTM D2440			
Pressure vessel test	minutes	ASTM D2112	195		252
Oxidation inhibitor content	wt %	ASTM D2668	0.08	0.3	0.28
Corrosive sulphur		ASTM D1275	Non-Corrosive		Non-Corrosive
Corrosive sulphur - Copper		ASTM D1275	Non-Corrosive		Non-Corrosive
Corrosive sulphur - Silver		ASTM D1275	Non-Corrosive		Non-Corrosive
Water content	mg/kg	ASTM D1533		35	< 20
Neutralization number	mg KOH/g	ASTM D974		0.03	< 0.01
Furanic compounds, per compound	µg/L	ASTM D5837		25	< 25
PCB	mg/kg	ASTM D4059	Not Detectable		Not Detectable

Nytro 11GBX-US is an inhibited insulating oil, meeting all requirements in the specification ASTM D3487-16, type II.

*As per definition given by ASTM D 3487-16 appendix X2.2.1.1, processing involves filtering, dehydration and degassing.

Severely hydrotreated insulating oil

Issuing date: 09-24-2020